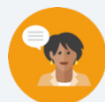


10.1 Graphing Absolute Value Functions

Lesson Introduction

 Benchmark	MA.912.AR.4.3 Given a table, equation, or written description of an absolute value function, graph that function and determine its key features.		 Learning Targets	- I can graph an absolute value function from a table or equation. - I can determine key features of an absolute value function.
 Benchmark Coherence	Building On N/A		Working Toward MA.912.AR.4.4 Solve and graph mathematical and real-world problems that are modeled with absolute value functions. Interpret key features and determine constraints in terms of the context.	
 Essential Questions	- How do I graph an absolute value function from a table? - How do I graph an absolute value function from an equation? - What key features can be determined from an absolute value function?			
 Lesson Narrative	In this lesson, students are introduced to absolute value functions. Students will begin by learning how to graph absolute value functions by using a table of values. They will then collaborate to identify key features of absolute value functions given a graph. Lastly, students will get the opportunity to receive guided practice with both graphing and identifying key features of absolute value functions.			
 Component Summary	10.1.1 Warm-Up (~5 min.)	Career exploration: Prototype Fabricator (<i>SE p. 121</i>)	10.1.3 Collaboration (15-20 min.)	Collaborative activity with identifying key features of absolute value functions (<i>SE p. 123</i>)
	10.1.2 Guided Instruction (10-15 min.)	Introduction to graphing absolute value functions (<i>SE p. 122</i>)	10.1.4 Guided Practice (15-20 min.)	Practice graphing and identifying key features of absolute value functions (<i>SE p. 124</i>)
	10.1.5 Wrap-Up (5 min.)	Concept self-reflection activity (<i>SE p. 125</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Graphing software to display the graphs separately for 10.1.3 Collaboration	10.1.1 Warm-Up video to accompany the question prompts

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may bring their misconceptions from graphing linear functions into graphing absolute value functions. - Students may plot the points incorrectly on the coordinate plane. - Students may misinterpret the key features from the graph.	10.1.3 Collaboration is an activity where students explore the key features of absolute value functions. This activity is designed to connect what students have learned about key features in prior units to what they just learned about graphing absolute value functions. Go through this activity prior to releasing to students. It will be important to let the activity unfold naturally, without giving away information they will gain naturally from the process.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share During 10.1.3 Collaboration, consider leveraging a Think-Pair-Share routine. Allow students to try the questions on their own, and then compare with their partner to discuss what they found and come to a consensus on the correct answer.	MA.K12.MTR.4.1: Discussion and Reflection In 10.1.3 Collaboration, students get to work together to explore the key features of absolute value functions given two different functions represented on the coordinate plane. During this activity, students will be positioned to communicate mathematical ideas, vocabulary, and methods effectively.
 Differentiate Instruction	Consider a variety of strategies for differentiating 10.1.3 Collaboration: - Assign each student to complete the work for one color with their partner, and then try the second color independently. - Have each student choose a color then compare their work and solutions. - Prompt students to be “experts” in certain concepts (intercepts or end behavior for example). Each student must then find partners in the classroom with a different ‘expertise’ and compare notes. Encourage students not to copy the work, but instead act as an expert teacher by explaining their area of expertise.	
Lesson Notes:		

10.1.1 Warm-Up: Prototype Fabricator

Component Overview: Students will explore the career of a prototype fabricator and use information from that exploration to answer two questions.



10.1.1 Warm-Up: Prototype Fabricator

1. Jeff Tiedeken says, “A lot of times people come in the door and they don’t even have an idea of what they need. They just have a problem. So, the first thing is to understand the problem.”



- a. Describe a time in any class where you first had to spend time to understand a problem.

Answers will vary.

- b. Why do you think, for Jeff Tiedeken, understanding the problem is the first thing to do?

Answers will vary.



Albert Einstein once said, “If I had one hour to solve a problem, I’d spend 55 minutes thinking about the problem, and 5 minutes thinking about the solutions.” What do Tiedeken’s and Einstein’s quotes have in common? What does that make you think about when solving problems in math?

10.1.2 Guided Instruction: Graphing Absolute Value Functions

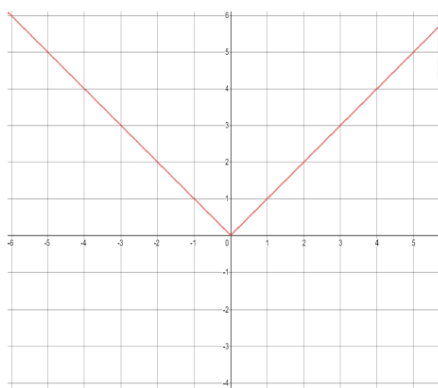
Component Overview: Students will be introduced to graphing absolute value functions using tables of values with three different absolute value function questions.



10.1.2 Guided Instruction: Graphing Absolute Value Functions

1. The table of values represents the absolute value function $f(x) = |x|$. Use the table of values to graph the function $f(x) = |x|$.

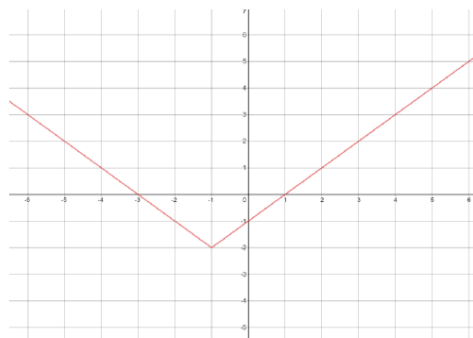
x	$f(x)$
-2	2
-1	1
0	0
1	1
2	2



What did you notice about the shape of the graph when you plotted all the points from the table? Why do you think this shape exists, based on what you know about absolute values?

2. The table of values represents the absolute value function $h(x) = |x + 1| - 2$. Use the table of values to graph the function $h(x) = |x + 1| - 2$.

x	$h(x)$
-3	0
-2	-1
-1	-2
0	-1
1	0



Notice and Wonder

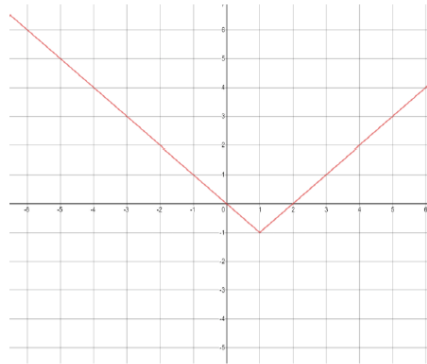
What do you notice about the values of the absolute value function? What does that make you wonder about the solutions of an absolute value function?

Students may describe what they see as having two “answers” (or x -values) for every unique y -value, leading to a conceptual understanding of solving algebraically.

10.1.2 Guided Instruction: Graphing Absolute Value Functions (continued)

3. Complete the table of values and graph the absolute value function $t(x) = |x - 1| - 1$.

x	$t(x)$
-3	3
-2	2
-1	1
0	0
1	-1
2	0
3	1



For question 3, show students how to complete the table using the absolute value function.

10.1.3 Collaboration: Identifying Key Features of Absolute Value Functions

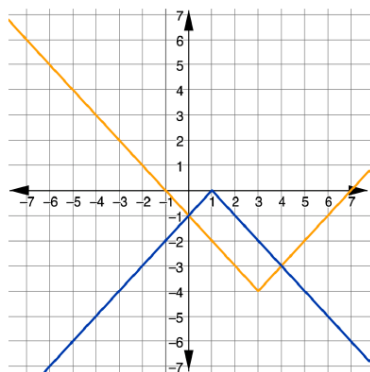
Component Overview: Students will work with a partner to identify the key features of two absolute value functions provided in graph form.



10.1.3 Collaboration: Identifying Key Features of Absolute Value Functions

Similar to quadratic functions, absolute value functions have a vertex and a line of symmetry.

Two absolute value functions are shown on the coordinate grid.



Work with your partner to complete the following.

- Complete the table for each function.

Function	Vertex	Equation of the Line of Symmetry
Blue	(1, 0)	$x = 1$
Orange	(3, -4)	$x = 3$

- List two points on the orange graph that are equidistant from the line of symmetry.
Answers may vary. (2, -3) and (4, -3)
- List two points on the blue graph that are equidistant from the line of symmetry.
Answers may vary. (0, -1) and (2, -1)



Some students may struggle with viewing these on the same graph as they look back and forth between the task in this activity and its corresponding graph. Consider making these two separate graphs using a graphing software for students to see them individually to engage in the collaboration independently.



Some students may have forgotten the meaning of the terms “line of symmetry” and “equidistant.” Consider vocabulary supports where needed, such as a word wall or anchor chart.

10.1.3 Collaboration: Identifying Key Features of Absolute Value Functions (continued)

4. Work with your partner to identify additional key features of the specified function.

Function	x -intercept(s)	y -intercept	End Behavior
Blue	(1,0)	(0,-1)	As $x \rightarrow \infty$, $y \rightarrow -\infty$. As $x \rightarrow -\infty$, $y \rightarrow -\infty$.
Orange	(-1,0) and (7,0)	(0,-1)	As $x \rightarrow \infty$, $y \rightarrow \infty$. As $x \rightarrow -\infty$, $y \rightarrow \infty$.

5. Complete the table for the orange function.

Key Feature	Inequality	Set-Builder Notation
Domain	$-\infty < x < \infty$	$\{x x \in R\}$
Range	$y \geq -4$	$\{y y \geq -4\}$
Increasing	$x > 3$	$\{x x > 3\}$
Decreasing	$x < 3$	$\{x x < 3\}$
Positive	$x < -1$ or $x > 7$	$\{x x < -1$ or $x > 7\}$
Negative	$-1 < x < 7$	$\{x -1 < x < 7\}$

6. Complete the table for the blue function.

Key Feature	Inequality	Set-Builder Notation
Domain	$-\infty < x < \infty$	$\{x x \in R\}$
Range	$y \leq 0$	$\{y y \leq 0\}$
Increasing	$x < 1$	$\{x x < 1\}$
Decreasing	$x > 1$	$\{x x > 1\}$
Positive	N/A	N/A
Negative	$x < 1$ or $x > 1$	$\{x x \neq 1\}$



While circulating during this activity, do a check on students' answers to see if they have the correct answers, but have associated them with the wrong functions. Students could either truly not understand, or just have looked at the wrong function.



What key features did you notice were the same for both the orange and blue functions?

What key features did you notice were different for the two functions?

10.1.4 Guided Practice: Graphing and Identifying Key Features of Absolute Value Functions

Component Overview: Students will practice both graphing and identifying key features of absolute value by answering five questions related to one absolute value function.

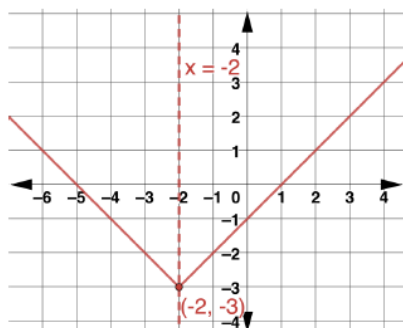


10.1.4 Guided Practice: Graphing and Identifying Key Features of Absolute Value Functions

Consider the function $g(x) = |x + 2| - 3$.

1. Complete the table of values to graph $g(x) = |x + 2| - 3$.

x	$g(x)$
-5	0
-4	-1
-3	-2
-2	-3
-1	-2
0	-1
1	0
2	1



2. Label the vertex and draw the line of symmetry on the graph.

Vertex = $(-2, -3)$

Line of Symmetry: $x = -2$

3. Complete the table.

x-intercept(s)	y-intercept
$(-5, 0)$ and $(1, 0)$	$(0, -1)$
Right End Behavior	Left End Behavior
$x \rightarrow \infty, y \rightarrow \infty$	$x \rightarrow -\infty, y \rightarrow \infty$

4. Use the inequality format to complete the table.

Domain	Range
$-\infty < x < \infty$	$y \geq -3$

5. Use set-builder notation to complete the table.

Interval(s) where the function is increasing	Interval(s) where the function is decreasing
$\{x x > -2\}$	$\{x x < -2\}$
Interval(s) where the function is positive	Interval(s) where the function is negative
$\{x x < -5 \text{ or } x > 1\}$	$\{x -5 < x < 1\}$



Consider how 10.1.4 Guided Practice might be an opportunity for informed differentiation in Lesson 2.

- Question 1 – Evaluate an absolute value expression and graph the resulting values accordingly.
- Question 2 – Location of the vertex and/or the line of symmetry (including appropriately writing the equation of the line of symmetry).
- Question 3 – Identify the x - and y -intercepts and/or describe the end behavior.
- Question 4 – Domain and range of an absolute value function.
- Question 5 – Increasing/decreasing and positive/negative intervals, as well as set-builder notation.

10.1.5 Wrap-Up: Reflection

Component Overview: Students will complete a self-reflection on their level of understanding of the concepts of this lesson.

- Circle the icon that best describes your understanding of each concept.

Answers will vary.



I need help.



I got it.



I can help someone.

graphing absolute value functions			
x -intercepts			
y -intercept			
vertex			
line of symmetry			
domain			
range			
increasing/decreasing			
positive/negative			
end behavior			



Compare student responses to their work in 10.1.4 Guided Practice. Alternatively, prompt students to identify which questions from 10.1.4 Guided Practice incorporate each concept as an additional layer of information for Lesson 2.